

Connectivity Diagnostics in a Pilot Generative Surrogate of Zero-Degree Calorimeter Showers

Julian Juan

September 2026

Abstract

We study a pilot generative model of single-neutron Geant4 showers in a 6,790-channel Zero-Degree Calorimeter readout. Conditioned on neutron energy and direction, it samples the total recorded deposit, allocates that deposit among active layers, selects active channels, and distributes each layer's energy among them. We compare one checkpoint trained on 26,624 events with Geant4 showers at 10,000 stored conditions in the 50–250 GeV incident kinetic-energy range. The mean total deposit differs by 0.045 GeV (1.0%); opposing shifts in the electromagnetic and hadronic sections partly cancel. Among nonempty events, mean active-layer counts differ by 0.25, but the mean last active layer is 4.42 layers farther downstream and the mean number of inactive layers within the occupied span is 3.95 higher. Mean active-channel counts differ by -28.7 (-1.8%), while the generated support has 35.97 more connected components on the model graph and a 6.03-percentage-point lower mean largest-component fraction. These descriptive differences concern one checkpoint, a strict-positive readout threshold, and a repeatedly inspected development sample. Event-level arrays needed for uncertainty estimates and threshold or graph sensitivity tests were not retained. The observed connectivity discrepancy has no established statistical significance or causal attribution.

1 Introduction

Detailed particle transport is central to detector design and high-energy-physics analysis, but calorimeter shower simulation is computationally expensive. Generative surrogates based on normalizing flows, diffusion, point clouds, graph networks, and conditional flow matching have consequently been developed for faster sample production [1–5]. Their evaluation must probe more than a single response summary. CaloChallenge compares one-dimensional observables, physics distances, classifier tests, model size, and generation time [6]; systematic studies of HEP generative metrics likewise show that different tests respond to different distributional failures [7].

Zero-Degree Calorimeters (ZDCs) are also the subject of dedicated surrogate studies. Work on the ALICE ZDC has considered variational autoencoders, generative adversarial networks, diffusion models, conditional flow matching, and mixtures of generative experts [8–11]. Those studies concern a different detector representation from the raw deposited-energy vector examined here, so their numerical results are not direct baselines for this checkpoint.

Spatial organization is physically relevant in calorimetry. For example, ATLAS topological clustering groups neighboring cells above signal-significance thresholds and uses the resulting cluster shapes in reconstruction [12]. This does not make every graph component a detector observable: connectivity depends on the readout mapping, the adjacency definition, and the energy threshold. It does motivate asking whether a surrogate that matches response and occupancy summaries also reproduces the organization of its occupied readout support.

We address that question for one pilot-trained checkpoint on an irregular 65-layer readout. Only aggregate results remain for the 10,000-event comparison, and this sample was inspected repeatedly during development. We therefore ask a narrow descriptive question: for these stored pairs, do similar mean

occupancies coexist with a different longitudinal reach or graph connectivity? The answer is yes under the model graph and the threshold $Y_i > 0$. The available record does not determine statistical significance, behavior after retraining, or the mechanism responsible for the difference.

2 Simulation target and evaluation population

For each event, the stored condition is an incident-neutron four-vector $p^\mu = (E, p_x, p_y, p_z)$. The model input is

$$c = (\log(1 + K_{\text{inc}}/100 \text{ GeV}), u_x, u_y, u_z, \log(E/\text{GeV})), \quad \mathbf{u} = \mathbf{p}/\|\mathbf{p}\|, \quad (1)$$

where $K_{\text{inc}} = E - m_n$. The two energy features are deterministically related by $E = K_{\text{inc}} + m_n$; retaining both is a frozen implementation choice and does not add independent conditioning information.

The target $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{6790}$ contains raw, nondigitized energy deposits summed by valid readout ID. It is not the incident-neutron energy or the total energy lost in all detector material. Layer 0 contains 400 electromagnetic-calorimeter (ECAL) channels. Hadronic-calorimeter (HCAL) layers 1–63 contain 100 channels each and layer 64 contains 90, for 6,390 HCAL channels. Of the HCAL readout IDs, 2,400 gang two to four stable physical positions. Each ganged channel is represented by the unweighted centroid of its positions (Fig. 1). The released metadata does not specify the full material composition, Geant4 version, physics list, production cuts, field configuration, or whether p^μ is recorded at generation or calorimeter entry. The observed few-GeV readout response can therefore be interpreted only as energy in the stored active-readout target, not as a calibrated calorimeter response fraction.

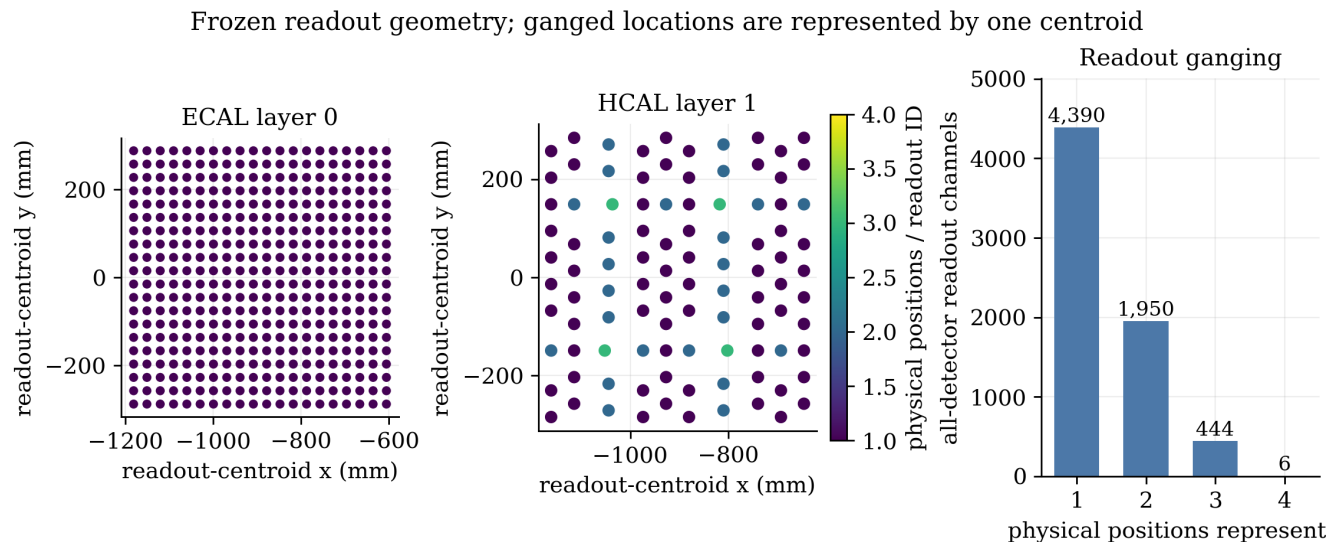


Figure 1: Frozen readout geometry. The left panels show ECAL layer 0 and HCAL layer 1. The ganging histogram covers all 6,790 readout channels; its first bar contains 400 ECAL and 3,990 HCAL channels.

The graph connects every channel to its eight nearest same-layer centroids. For each node in layer $\ell = 0, \dots, 63$, four nearest centroids are selected in layer $\ell + 1$, and each selected longitudinal pair is stored in both directions. This gives $8 \times 6790 = 54,320$ lateral directed edges and $2 \times 4 \times (400 + 63 \times 100) = 53,600$ longitudinal directed edges, or 107,920 in total. The same graph is used by the generator and by the connectivity diagnostic. It is an implementation graph, not a uniquely established physical-neighbor graph, and centroiding removes separation within a ganged channel.

The preparation record contains 764,940 events over 0–300 GeV, divided by deterministic event hash into 612,482 training, 76,158 validation, and 76,300 nominal test events. The checkpoint studied here used a smaller pilot of 26,624 training events and 6,656 validation events. Its training population is 4.35% of the canonical training partition.

All shower comparisons use a 10,000-condition sample from the canonical validation partition with $50 \leq K_{\text{inc}} \leq 250 \text{ GeV}$. We call this sample D_{dev} : it was inspected during model and manuscript development, and its overlap with the 6,656-event pilot-validation sample is not recoverable from the released aggregates. No nominal test event is used in this manuscript. The results are therefore descriptive measurements of the stored pairs, rather than an untouched assessment of expected model performance.

3 Generator and training procedure

The generator has 2,082,507 trainable parameters. It receives the five conditioning features in Eq. (1); no reference shower is supplied when generating an event. A two-block residual multilayer perceptron maps these inputs to a 128-component representation h_c , which conditions all subsequent stages. Here a *head* is a neural network that predicts the parameters of a distribution from which a quantity is drawn. A *conditional flow* starts from Gaussian random numbers and follows a learned velocity field to produce correlated continuous outputs. The model has no learned shower encoder or common random latent vector: independent draws enter at the discrete heads and at the two flows.

Figure 2 follows the sampling path. Each draw conditions later stages. Thus a change in total deposit can affect every later allocation, whereas the final channel-energy assignment cannot change a previously sampled layer budget or channel count.

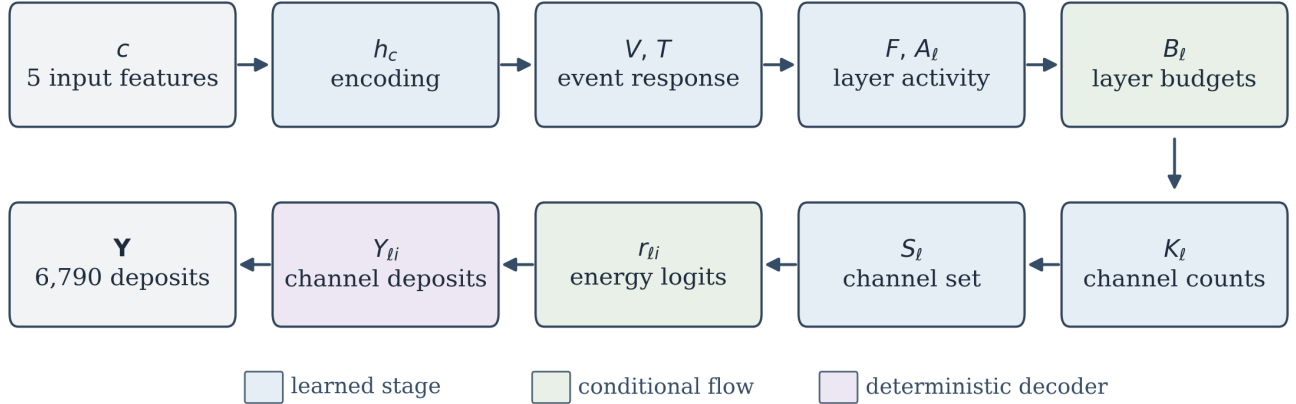


Figure 2: Event generation runs left to right across the top row, then right to left across the bottom row. Blue boxes are learned conditioning and sampling stages, green boxes are conditional flows, and the purple box is the deterministic decoder. V indicates any positive readout deposit, T is its total, F is the first active layer, A_ℓ is layer activity, B_ℓ is the layer budget, K_ℓ is the active-channel count, and S_ℓ is the selected channel set. The flow output $r_{\ell i}$ gives channel-energy logits, normalized within each layer to obtain $Y_{\ell i}$. During training, stages receive upstream quantities derived from the reference shower; generation uses earlier sampled outputs.

3.1 Total deposit and longitudinal allocation

A Bernoulli head samples V , which indicates whether the event has any positive readout deposit. A second head outputs the weights, means, and positive widths of a four-component Gaussian mixture for the transformed total deposit

$$q_T = \log(1 + T/10 \text{ GeV}). \quad (2)$$

When $V = 1$, the implementation samples a mixture component and a normal variate, applies the inverse transform, and sets a negative result to zero. It then limits the result to

$$0 \leq T \leq C(K_{\text{inc}}), \quad C(K_{\text{inc}}) = \min\{0.630110 \max(K_{\text{inc}}, 1 \text{ GeV}), 61.2338 \text{ GeV}\}. \quad (3)$$

The cap is a numerical bound derived from the training sample. If $V = 0$, or if inversion and clipping give $T = 0$, no layer is activated. The aggregate evaluation did not retain cap-activation or reference-exceedance counts, so the response tail near this boundary is untested.

For $V = 1$, a categorical head samples the first active layer F from (h_c, T) . Layers before F are set inactive, layer F is forced active, and a Bernoulli head samples the later activity indicators. The later indicators are independent conditional on the variables supplied to that head,

$$p(\mathbf{A} \mid h_c, T, F) = \prod_{\ell > F}^{64} p(A_\ell \mid h_c, T, F, \ell), \quad A_F = 1. \quad (4)$$

This head does not require the active layers to be contiguous.

The first conditional flow apportions T among active layers. For training, each reference layer deposit is divided by the reference total; the logarithms of these fractions, floored at 10^{-8} , are centered over active layers to form a target vector x_1 . The flow draws a standard-normal vector x_0 and a time t uniformly in $[0, 1]$, then constructs

$$x_t = (1 - t)x_0 + tx_1, \quad \mathcal{L}_{\text{CFM}} = \|v_\theta(x_t, t; \text{context}) - (x_1 - x_0)\|_2^2, \quad (5)$$

The squared error trains the velocity field v_θ to point along the straight path from x_0 to x_1 [13]. At generation, the field advances a fresh Gaussian vector through eight fixed updates, with velocity evaluated at the midpoint time of each interval. A softmax over active layers converts its output to nonnegative fractions π_ℓ that sum to one. The layer budgets are

$$B_\ell = T\pi_\ell, \quad B_\ell = 0 \text{ when } A_\ell = 0, \quad \sum_{\ell=0}^{64} B_\ell = T. \quad (6)$$

The velocity field uses a 128-component layer representation and two bidirectional transformer layers with four attention heads. Its context comprises h_c , T , the sampled activity mask, layer identity, and a sinusoidal encoding of flow time.

3.2 Channel count, support, and deposited energy

For each layer, a categorical head receives (h_c, B_ℓ, A_ℓ) and a learned layer identifier, then samples the active-channel count K_ℓ . Its feasibility mask permits $K_\ell = 0$ in an inactive layer and $1 \leq K_\ell \leq N_\ell$ in an active layer, where N_ℓ is that layer’s channel count. The mask fixes a count but imposes no minimum energy per channel in this zero-threshold study.

The support network scores every channel for selection. Its inputs for each channel are standardized (x, y, z) coordinates, normalized layer index, ECAL/HCAL indicators, h_c , the layer budget B_ℓ , and the requested occupancy fraction K_ℓ/N_ℓ . Three 96-component graph blocks update each channel representation from its neighbors. Their five edge inputs specify coordinate differences, distance, and whether an edge is lateral, forward-longitudinal, or backward-longitudinal. Channel representations are averaged within each layer; a two-layer, four-head bidirectional transformer exchanges information among the 65 layer summaries; the resulting context is returned to each channel. Independent Gumbel noise is added to the channel scores, and the K_ℓ highest scores select exactly K_ℓ channels in layer ℓ [14]. Selection has no constraint that would make those channels a connected set.

The second conditional flow divides each sampled layer budget among its selected channels. Its training target is obtained by taking the log of each reference channel’s fraction of its layer energy, flooring fractions at 10^{-8} , and centering the logs within that layer. It uses the same straight-path objective as Eq. (5). Its velocity network has the support network’s graph and layer-context structure, with the current flow state and a sinusoidal time encoding added at each channel. Unselected channels are masked before and after

every graph block. At generation, independent Gaussian values on selected channels undergo eight fixed updates to produce logits $r_{\ell i}$. The deterministic decoder computes

$$Y_{\ell i} = \begin{cases} B_{\ell} \frac{\exp r_{\ell i}}{\sum_{j \in S_{\ell}} \exp r_{\ell j}}, & i \in S_{\ell}, \\ 0, & i \notin S_{\ell}, \end{cases} \quad (7)$$

where S_{ℓ} is the selected set. Thus, in exact arithmetic, $|S_{\ell}| = K_{\ell}$, $Y_{\ell i} \geq 0$, $\sum_i Y_{\ell i} = B_{\ell}$, and $\sum_{\ell i} Y_{\ell i} = T$. These identities conserve the model’s sampled readout budget. They do not enforce conservation of incident-neutron energy, agreement with Geant4, or connectivity of the selected support.

3.3 Training and checkpoint selection

Training derives V , T , F , A_{ℓ} , B_{ℓ} , K_{ℓ} , S_{ℓ} , and both flow targets from each reference shower. A component receives the reference values of earlier components, whereas generation passes its own sampled values downstream. This training procedure is often called teacher forcing. The nine losses are binary cross entropy for visibility and activity, mixture negative log likelihood for T , categorical cross entropy for F and K_{ℓ} , binary cross entropy plus pairwise ranking for support, and mean-squared velocity error for each flow. Weights were obtained from gradient norms measured on training data. The response, layer-profile, and count weights reached the lower clipping bound, and the visibility weight reached the upper bound. These weights describe the optimization; they do not identify the source of the connectivity discrepancy.

The selected joint-training run used AdamW with a peak learning rate of 3×10^{-4} , batch size six, accumulation over four batches, gradient clipping at one, and seed 20260723. Epoch 90 has the lowest recorded validation objective among the 104 retained rows spanning epochs 11–114 and defines the checkpoint studied here. Validation objectives are lower than training objectives over much of this record. The retained aggregates do not identify why, so the ordering is not given a physical interpretation. Both flows use eight integration updates at evaluation; no solver-step comparison was retained.

4 Diagnostics

We compare mean total, ECAL, and HCAL deposits over all events. For eight 25 GeV incident-energy bins, retained means and standard deviations describe how these deposits vary with energy. The report contains no binwise uncertainty estimates.

A layer or channel is active when its stored energy is strictly positive; no detector or analysis threshold is applied. For a nonempty event with first and last active layers F and L , the occupied span is $L - F + 1$, and the number of inactive layers inside it is

$$G = L - F + 1 - \sum_{\ell=0}^{64} A_{\ell}. \quad (8)$$

This counts inactive layers, not contiguous runs. A shower has an interior gap when $G > 0$. Ignoring edge direction, the occupied channels S form connected components $C_1, \dots, C_m$ on the model graph. We report their number m and the fraction of occupied channels in the largest one, $\max_j |C_j|/|S|$. These support summaries use only nonempty events: 9,907 reference and 9,858 generated events.

The retained condition-only pipeline control has AUROC 0.500. Because it contains no shower variables, it checks only that identical condition distributions are presented to the two labels; it is not a two-sample test of the generated showers. A valid classifier test would require held-out discrimination [7] and, for this paired design, a split that keeps each matched reference–generated pair together.

5 Results

Table 1 reports absolute differences so that ratios on small denominators do not exaggerate the comparison. The all-event mean total deposit differs by 0.045 GeV, but this sum combines an ECAL excess of 0.175 GeV with an HCAL deficit of 0.129 GeV. Across the eight incident-energy bins, the generated-minus-reference mean difference ranges from -7.7% to $+7.7\%$, while the standard-deviation difference ranges from -10.9% to $+9.6\%$. Without binwise uncertainties these ranges diagnose possible cancellation but do not establish a statistically resolved energy dependence.

Table 1: Aggregate development-bank results. Deposit rows and zero-deposit counts use all 10,000 pairs. Support rows are conditional on nonempty events (9,907 reference and 9,858 generated). Percentage-point differences are denoted pp. No uncertainty intervals can be reconstructed from the retained aggregates.

Quantity	Geant4 reference	Generator	Gen. – ref.
Zero-deposit events	93 (0.93%)	142 (1.42%)	+49 (+0.49 pp)
Mean total deposit (GeV)	4.324	4.369	+0.045
Mean ECAL deposit (GeV)	2.100	2.275	+0.175
Mean HCAL deposit (GeV)	2.224	2.094	−0.129
Mean first active layer	0.510	0.735	+0.225
Mean last active layer	56.20	60.62	+4.42
Mean active layers	54.58	54.83	+0.25
Mean occupied span	56.69	60.88	+4.20
Mean inactive layers in span	2.10	6.05	+3.95
Events with an interior gap	52.43%	93.53%	+41.10 pp
Mean active channels	1617.6	1588.8	−28.7
Mean weak graph components	23.42	59.39	+35.97
Mean largest-component fraction	94.90%	88.87%	−6.03 pp

The longitudinal discrepancy is more directly described as a shift in reach than as a ratio of gap counts. The mean active-layer counts are close, but the generated sample’s mean first active layer is 0.225 layer later and its mean last active layer is 4.42 layers later. The mean occupied span consequently grows by 4.20 layers while the mean active-layer count grows by only 0.25, leaving 3.95 additional inactive layers inside the span on average. Figure 3 shows the corresponding mean energy allocation.

At channel level, the generated sample has 28.7 fewer active channels and 35.97 more weak components on average. Its mean largest-component fraction is lower by 6.03 percentage points. Under the declared graph and strict-positive definition, the generated support is therefore more disconnected even though its mean cardinality is similar. The available aggregates do not record component energies. They cannot determine whether the additional components carry substantial energy or arise from small positive deposits produced by the selected-channel softmax.

There are 49 more zero-deposit generated events than reference events. A significance calculation would require the joint paired zero/nonzero counts, which were not retained; the marginal counts alone do not support one. The result is reported because the visibility hurdle is part of the sampled model and because excluding empty events changes the denominators of every support statistic.

The decoder-level checks passed for all 1,250 batches of eight generated events. No nonfinite value, negative value, invalid support, requested-versus-realized count mismatch, or positive-support mismatch was recorded. Maximum event and layer budget residuals were 1.14×10^{-5} and 3.05×10^{-5} GeV under the evaluator’s batch-wide absolute-plus-relative tolerance. These checks establish consistency with sampled budgets and counts for this generated bank; they do not test agreement with Geant4.

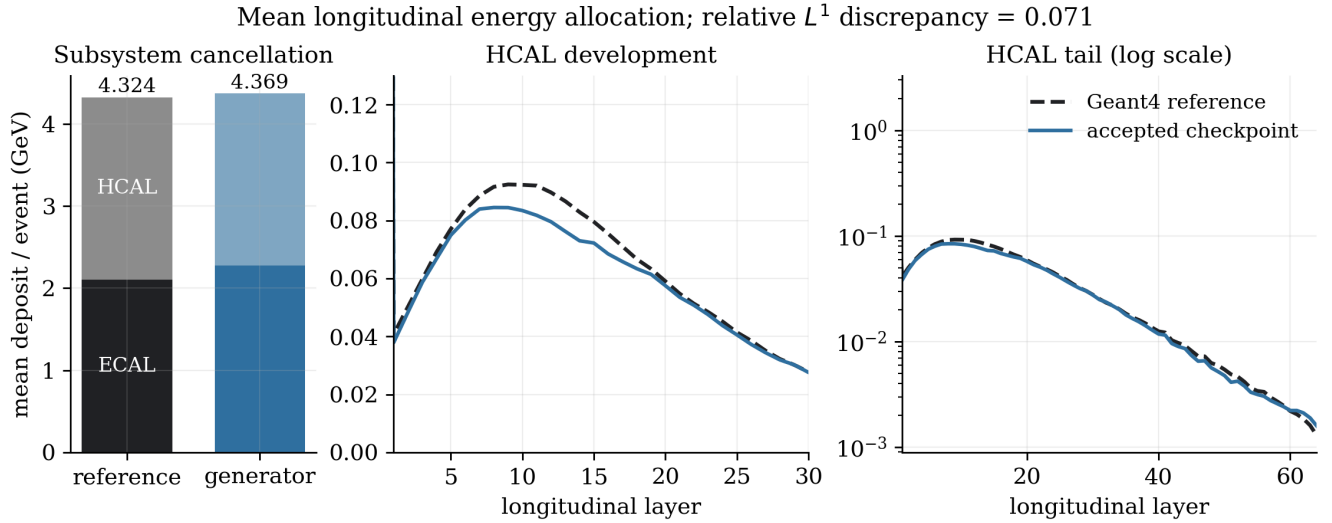


Figure 3: Mean longitudinal allocation on D_{dev} . The stacked bars expose the opposing ECAL and HCAL shifts. The layer panels show HCAL layers only; ECAL layer 0 is isolated in the bars. The curves are descriptive means without event-level uncertainty bands.

6 Discussion

The empirical statement supported by these aggregates is specific: one pilot checkpoint produces a later and more disconnected strict-positive readout support than its paired Geant4 sample while retaining similar mean layer and channel multiplicities. Exact cardinality and budget allocation leave the spatial arrangement underdetermined, so this coexistence is compatible with the decoder construction. The comparison does not show that the factorization caused the discrepancy. Limited training data, checkpoint selection, numerical integration, the shared graph, and the zero threshold remain alternative explanations.

Connectivity is one useful projection of shower structure, but it is not a complete or representation-independent fidelity metric. CaloChallenge and CaloDREAM illustrate the broader practice of combining one-dimensional shower observables with multivariate diagnostics [5, 6]. The HEP metric literature similarly recommends complementary tests rather than a single scalar ranking [7]. For this readout, a defensible connectivity result would additionally require thresholds tied to detector noise or digitization, energy-weighted component summaries, alternative adjacency graphs, and distributions resolved in incident energy and direction.

Several additional tests are needed because they answer different questions. Repeating generation with several sampling seeds would measure Monte Carlo variability for the fixed checkpoint. Neighboring checkpoints and independent training seeds would test selection and training variability. A 26k–100k–full-data scaling study would test whether the result changes with training-set size. Replacing generated upstream variables with their reference counterparts in an oracle cascade would show where the discrepancy first appears in the hierarchy. Solver-step scans would test numerical integration. Matched alternatives for layer activity and support selection would test whether those design choices alter connectivity. The present aggregate file cannot supply any of these comparisons.

D_{dev} was repeatedly inspected, its overlap with pilot validation is unresolved, and event-level arrays were not retained. A physics-performance paper would require a locked evaluation bank with documented identities, paired uncertainty estimates, full structural distributions, a valid shower-aware classifier test, reconstructed-observable comparisons, and end-to-end timing. A complete reference-simulation specification and the applicable collaboration approval would also be required. The present evidence therefore supports an exploratory computational-physics case study, but not a validated fast-simulation result.

7 Conclusion

For the single pilot checkpoint studied here, close mean occupancies do not coincide with close readout-support organization. Relative to the paired Geant4 sample, generated events have 0.25 more active layers on average but a mean last active layer 4.42 layers farther downstream, leaving 3.95 more inactive layers within the occupied span on average. They have 28.7 fewer active channels and 35.97 more weak components on average, with a 6.03-percentage-point reduction in the mean largest-component fraction. Opposing ECAL and HCAL shifts also partially cancel in the mean total deposit.

These values describe one repeatedly inspected development bank under a strict-positive threshold and one shared graph. They document a discrepancy in the retained aggregate report. They do not establish its statistical significance, physical relevance after digitization, persistence across model fits, or causal origin. Those questions require the event-level, threshold, graph, seed, and locked-evaluation studies specified above.

Code and data availability

The manuscript source, aggregate validation report, event-independent geometry summary, training-history extract, figure builder, and QA records are available at <https://github.com/JulianAttemptsCoding/fast-mc-paper>. The collaboration-owned event file and checkpoint are not redistributed. The public package reproduces the manuscript and aggregate arithmetic; it cannot retrain the model, regenerate showers, or reconstruct event-level tests.

Acknowledgments

The author thanks Dr. Wen-Chen Chang (Institute of Physics, Academia Sinica) for mentorship and guidance on the experimental high-energy-physics context of this project.

Author contributions

Julian Juan: conceptualization, methodology, software, investigation, visualization, and writing.

Use of generative AI

Generative-AI tools were used for editorial assistance, code generation, literature discovery, and adversarial manuscript auditing. Numerical statements were checked against versioned project artifacts during this revision; the author retains responsibility for verifying and approving the final manuscript.

References

- [1] C. Krause and D. Shih. “Fast and Accurate Simulations of Calorimeter Showers with Normalizing Flows”. In: *Physical Review D* 107 (2023), p. 113003. DOI: [10.1103/PhysRevD.107.113003](https://doi.org/10.1103/PhysRevD.107.113003). arXiv: [2106.05285](https://arxiv.org/abs/2106.05285).
- [2] O. Amram and K. Pedro. “Denoising Diffusion Models with Geometry Adaptation for High Fidelity Calorimeter Simulation”. In: *Physical Review D* 108 (2023), p. 072014. DOI: [10.1103/PhysRevD.108.072014](https://doi.org/10.1103/PhysRevD.108.072014). arXiv: [2308.03876](https://arxiv.org/abs/2308.03876).
- [3] D. Kobylanskii, N. Soybelman, E. Dreyer, and E. Gross. “Graph-Based Diffusion Model for Fast Shower Generation in Calorimeters with Irregular Geometry”. In: *Physical Review D* 110 (2024), p. 072003. DOI: [10.1103/PhysRevD.110.072003](https://doi.org/10.1103/PhysRevD.110.072003). arXiv: [2402.11575](https://arxiv.org/abs/2402.11575).
- [4] E. Buhmann et al. “CaloClouds II: Ultra-Fast Geometry-Independent Highly-Granular Calorimeter Simulation”. In: *Journal of Instrumentation* 19.04 (2024), P04020. DOI: [10.1088/1748-0221/19/04/P04020](https://doi.org/10.1088/1748-0221/19/04/P04020). arXiv: [2309.05704](https://arxiv.org/abs/2309.05704) [[physics.ins-det](https://arxiv.org/archive/physics)].

- [5] L. Favaro, A. Ore, S. Palacios Schweitzer, and T. Plehn. “CaloDREAM: Detector Response Emulation via Attentive Flow Matching”. In: *SciPost Physics* 18 (2025), p. 088. DOI: [10.21468/SciPostPhys.18.3.088](https://doi.org/10.21468/SciPostPhys.18.3.088). arXiv: [2405.09629](https://arxiv.org/abs/2405.09629).
- [6] C. Krause et al. “CaloChallenge 2022: A Community Challenge for Fast Calorimeter Simulation”. In: *arXiv preprint* (2024). arXiv: [2410.21611v1](https://arxiv.org/abs/2410.21611v1) [[physics.ins-det](#)].
- [7] R. Kansal et al. “Evaluating Generative Models in High Energy Physics”. In: *Physical Review D* 107 (2023), p. 076017. DOI: [10.1103/PhysRevD.107.076017](https://doi.org/10.1103/PhysRevD.107.076017). arXiv: [2211.10295](https://arxiv.org/abs/2211.10295) [[hep-ex](#)].
- [8] J. Dubiński, K. Deja, S. Wenzel, P. Rokita, and T. Trzciński. “Machine Learning Methods for Simulating Particle Response in the Zero Degree Calorimeter at the ALICE Experiment, CERN”. In: *AIP Conference Proceedings* 3061 (2024), p. 040001. DOI: [10.1063/5.0203567](https://doi.org/10.1063/5.0203567). arXiv: [2306.13606](https://arxiv.org/abs/2306.13606) [[cs.CV](#)].
- [9] M. Kita, J. Dubiński, P. Rokita, and K. Deja. *Generative Diffusion Models for Fast Simulations of Particle Collisions at CERN*. 2024. arXiv: [2406.03233](https://arxiv.org/abs/2406.03233) [[physics.data-an](#)].
- [10] M. Wojnar. *Even Faster Simulations with Flow Matching: A Study of Zero Degree Calorimeter Responses*. 2025. arXiv: [2507.18811](https://arxiv.org/abs/2507.18811).
- [11] P. Będkowski, J. Dubiński, F. Szatkowski, K. Deja, P. Rokita, and T. Trzciński. *ExpertSim: Fast Particle Detector Simulation Using Mixture-of-Generative-Experts*. 2025. arXiv: [2508.20991](https://arxiv.org/abs/2508.20991) [[cs.CV](#)].
- [12] ATLAS Collaboration. “Topological Cell Clustering in the ATLAS Calorimeters and Its Performance in LHC Run 1”. In: *European Physical Journal C* 77 (2017), p. 490. DOI: [10.1140/epjc/s10052-017-5004-5](https://doi.org/10.1140/epjc/s10052-017-5004-5). arXiv: [1603.02934](https://arxiv.org/abs/1603.02934) [[hep-ex](#)].
- [13] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. “Flow Matching for Generative Modeling”. In: *International Conference on Learning Representations*. 2023. arXiv: [2210.02747](https://arxiv.org/abs/2210.02747).
- [14] W. Kool, H. van Hoof, and M. Welling. “Stochastic Beams and Where to Find Them: The Gumbel-Top- k Trick for Sampling Sequences Without Replacement”. In: *Proceedings of the 36th International Conference on Machine Learning*. Vol. 97. Proceedings of Machine Learning Research. 2019, pp. 3499–3508.